IRREDUCIBLE COMPONENTS OF THE LIMIT AND THE COLIMIT OF RESIDUE FIELDS EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Topology of affine limits. Limits, topology lemmas of the chapter

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CUE: sources/stacks-chow/tags.txt, source lines 16099–16099. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $S = \lim S_i$ be the limit of a directed inverse system of schemes with affine transition morphisms (Lemma 01YX). Let $s \in S$ with images $s_i \in S_i$. Then*

(1) *$s = \lim s_i$ as schemes, i.e., $\kappa(s) = \operatorname{colim} \kappa(s_i)$,*
(2) *$\overline{\{s\}} = \lim \overline{\{s_i\}}$ as sets, and*
(3) *$\overline{\{s\}} = \lim \overline{\{s_i\}}$ as schemes where $\overline{\{s\}}$ and $\overline{\{s_i\}}$ are endowed with the reduced induced scheme structure.*

Proof. Choose $0 \in I$ and an affine open covering $S_0 = \bigcup_{j \in J} U_{0,j}$. For $i \geq 0$ let $U_{i,j} = f_{i,0}^{-1}(U_{0,j})$ and set $U_j = f_0^{-1}(U_{0,j})$. Here $f_{i'i} : S_{i'} \to S_i$ is the transition morphism and $f_i : S \to S_i$ is the projection. For $j \in J$ the following are equivalent: (a) $s \in U_j$, (b) $s_0 \in U_{0,j}$, (c) $s_i \in U_{i,j}$ for all $i \geq 0$. Let $J' \subset J$ be the set of indices for which (a), (b), (c) are true. Then $\overline{\{s\}} = \bigcup_{j \in J'} (\overline{\{s\}} \cap U_j)$ and similarly for $\overline{\{s_i\}}$ for $i \geq 0$. Note that $\overline{\{s\}} \cap U_j$ is the closure of the set $\{s\}$ in the topological space U_j. Similarly for $\overline{\{s_i\}} \cap U_{i,j}$ for $i \geq 0$. Hence it suffices to prove the lemma in the case S and S_i affine for all i. This reduces us to the algebra question considered in the next paragraph.

Suppose given a system of rings $(A_i, \varphi_{ii'})$ over I. Set $A = \operatorname{colim}_i A_i$ with canonical maps $\varphi_i : A_i \to A$. Let $\mathfrak{p} \subset A$ be a prime and set $\mathfrak{p}_i = \varphi_i^{-1}(\mathfrak{p})$. Then

$$V(\mathfrak{p}) = \lim_i V(\mathfrak{p}_i)$$

This follows from Lemma 0CUE because $A/\mathfrak{p} = \operatorname{colim} A_i/\mathfrak{p}_i$. This equality of rings also shows the final statement about reduced induced scheme structures holds true. The equality $\kappa(\mathfrak{p}) = \operatorname{colim} \kappa(\mathfrak{p}_i)$ follows from the statement as well. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.